



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 6

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Download any dataset (without considering the information of class label) from the UC Irvine Machine Learning Repository. Write a program to do the following:
 - (a) Read the dataset.
 - (b) Cluster the dataset into different groups using K-means clustering algorithm (**with and without using in-built Python library/ packages**) Consider the following for this:
 - Number of clusters, K : 2 to 8.
Hint: Apply K-means clustering, separately for all the given K -values. Use Elbow method to determine a suitable value of K .
 - **Distance measure:** Euclidean distance
 - (c) Report and visualize the obtained final clusters (including Silhouette score) with respect to different K , and write the result analysis.
2. Download any dataset (without considering the information of class label) from the UC Irvine Machine Learning Repository. Write a program to do the following:
 - (a) Read the dataset.
 - (b) Cluster the dataset using DBSCAN Clustering method (**with and without using in-built Python library/ packages**) considering the different values of algorithmic parameters.
 - (c) Report and visualize the obtained clusters and write the result analysis.